EFL Championship Promotion & Points Analysis

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```
#importing libraries
library(tidyverse)
library(tidymodels)
library(ggplot2)
library(naniar)
library(patchwork)
library(MASS)
library(car)
library(GGally)
library(bigrquery)
library(DBI)
library(gt)
library(dplyr)
library(skimr)
library(broom)
library(purrr)
library(gt)
library(ggrepel)
library(car)
```

1. Data Import and Cleaning

```
# Import data from BigQuery
project_id <- "eplpromo"
dataset_id <- "championship_data"

con <- dbConnect(
  bigrquery::bigquery(),
```

```
  project = project_id,
  dataset = dataset_id,
  billing = project_id
)
efl_table <- dbReadTable(con, "efl_summary")

# Load reference dataset
efl_table_og <- read_csv("EFL OG.csv")

# Clean dataset: Exclude incomplete 2024/25 season and retain RAW Shot_Accuracy (NO Standardi
efl_table_final <- efl_table |>
  dplyr::select(Season, Team, Total_Points, GF, GA, GD, Shot_Accuracy, Net_SOT, Card_Points,
  filter(Season != "2024/25") |>
  mutate(Promoted = factor(ifelse(Promoted == 1, "Yes", "No"), levels = c("Yes", "No")))

# Data Completeness Check
skim(efl_table_final)
```

Table 1: Data summary

Name	efl_table_final
Number of rows	480
Number of columns	10
_____	
Column type frequency:	
character	2
factor	1
numeric	7
_____	
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
Season	0	1	7	7	0	20	0
Team	0	1	3	18	0	56	0

Variable type: factor

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
Promoted	0	1	FALSE	2	No: 420, Yes: 60

Variable type: numeric

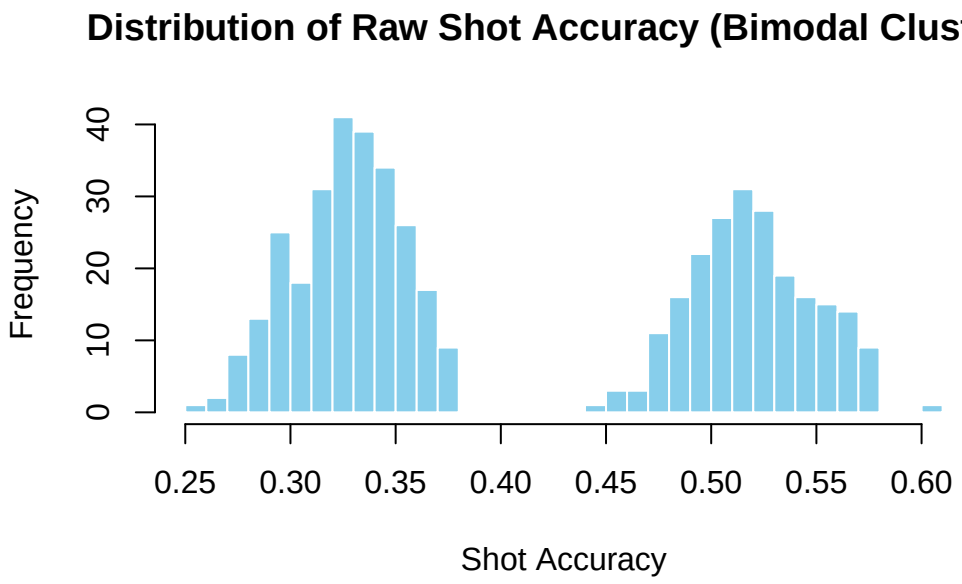
skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
Total_Points	0	1	62.70	14.60	23.00	53.00	61.00	72.00	106.0	
GF	0	1	58.79	12.65	29.00	50.00	58.00	66.00	106.0	
GA	0	1	58.79	11.86	30.00	51.00	59.00	67.00	98.0	
GD	0	1	0.00	20.16	-58.00	-13.00	-1.00	11.25	67.0	
Shot_Accuracy	0	1	0.41	0.10	0.25	0.33	0.36	0.52	0.6	
Net_SOT	0	1	0.00	51.87	-186.00	-36.00	-2.00	34.25	174.0	
Card_Points	0	1	83.62	15.74	28.00	73.00	84.00	95.00	130.0	

```
#The data turns out to have no missing values.
```

2. Structural Analysis of Shot Accuracy (Cloud Investigation)

Raw `Shot_Accuracy` exhibits a distinct bimodal distribution divided into two separate "clouds" (Lower Cloud vs. Higher Cloud) due to historical shifts in data collection criteria across eras. Rather than standardizing the variable, we analyze the two regimes explicitly using raw values.

```
# Histogram of Raw Shot Accuracy demonstrating bimodal clustering
hist(
  efl_table_final$Shot_Accuracy,
  main = "Distribution of Raw Shot Accuracy (Bimodal Clusters)",
  xlab = "Shot Accuracy", col = "skyblue", border = "white", breaks = 30
)
```




```
# Classify observations into Lower Cloud (< 0.35) and Higher Cloud (>= 0.35)
efl_table_final <- efl_table_final |>
  mutate(Cloud_Group = if_else(Shot_Accuracy < 0.35, "Lower Cloud", "Higher Cloud"))

# Summary Table 1: Overall Frequency and Appearance Counts of Clouds
cloud_summary <- efl_table_final |>
  group_by(Cloud_Group) |>
  summarise(
    Appearance_Count = n(),
    Percentage = round((n() / nrow(efl_table_final)) * 100, 2),
    Mean_Accuracy = round(mean(Shot_Accuracy, na.rm = TRUE), 4),
    Min_Accuracy = round(min(Shot_Accuracy, na.rm = TRUE), 4),
    Max_Accuracy = round(max(Shot_Accuracy, na.rm = TRUE), 4)
  )

cloud_summary |>
  gt() |>
  tab_header(
    title = md("**Shot Accuracy Cloud Frequency Summary**"),
    subtitle = "Quantifying Appearance Counts Across Data Regimes"
  ) |>
  cols_label(
    Cloud_Group = "Cloud Regime",
```

Shot Accuracy Cloud Frequency Summary

Quantifying Appearance Counts Across Data Regimes

Cloud Regime	Frequency (N)	Proportion (%)	Mean	Min	Max
Higher Cloud	268	55.83	0.4893	0.3501	0.6016
Lower Cloud	212	44.17	0.3181	0.2500	0.3498

```
    Appearance_Count = "Frequency (N)",
    Percentage = "Proportion (%)",
    Mean_Accuracy = "Mean",
    Min_Accuracy = "Min",
    Max_Accuracy = "Max"
  )
```

```
#Why did the shot accuracy have a disparity between two groups?
#The way that Opta defines shot accuracy has changed since 2013/14 season. Since 2014, there

#The histogram on the left-side shows the data since 2014.

#The summary table shows the number of teams that had lower/higher frequency, and their propo
```

```
# Summary Table 2: Seasonal Cloud Shift Breakdown
season_cloud_summary <- efl_table_final |>
  group_by(Season) |>
  summarise(
    Team_Count = n(),
    Avg_Shot_Accuracy = mean(Shot_Accuracy, na.rm = TRUE),
    Dominant_Cloud = ifelse(mean(Shot_Accuracy, na.rm = TRUE) < 0.35, "Lower Cloud", "Higher
  ) |>
  arrange(Season)

season_cloud_summary |>
  gt() |>
  tab_header(
    title = md("**Seasonal Breakdown of Shot Accuracy Regime**")
  ) |>
  cols_label(
    Season = "Season",
    Team_Count = "Teams (N)",
    Avg_Shot_Accuracy = "Mean Shot Accuracy",
```

Seasonal Breakdown of Shot Accuracy Regime

Season	Teams (N)	Mean Shot Accuracy	Dominant Regime
2004/05	24	0.534	Higher Cloud
2005/06	24	0.513	Higher Cloud
2006/07	24	0.505	Higher Cloud
2007/08	24	0.511	Higher Cloud
2008/09	24	0.527	Higher Cloud
2009/10	24	0.529	Higher Cloud
2010/11	24	0.524	Higher Cloud
2011/12	24	0.516	Higher Cloud
2012/13	24	0.521	Higher Cloud
2013/14	24	0.325	Lower Cloud
2014/15	24	0.322	Lower Cloud
2015/16	24	0.324	Lower Cloud
2016/17	24	0.327	Lower Cloud
2017/18	24	0.322	Lower Cloud
2018/19	24	0.328	Lower Cloud
2019/20	24	0.323	Lower Cloud
2020/21	24	0.321	Lower Cloud
2021/22	24	0.341	Lower Cloud
2022/23	24	0.325	Lower Cloud
2023/24	24	0.335	Lower Cloud

```
    Dominant_Cloud = "Dominant Regime"
  ) |>
  fmt_number(columns = Avg_Shot_Accuracy, decimals = 3)
```

The table above shows that if the shot accuracy is lower than 0.35, then the season belongs to the lower cloud of shot accuracy.

```
# Summary Table 3: Team-Level Shot Accuracy, Seasons, and Appearance Frequencies
team_shot_summary <- efl_table_final |>
  group_by(Team, Cloud_Group) |>
  summarise(
    Appearances = n(),
    Seasons = paste(Season, collapse = ", "),
    Mean_Shot_Accuracy = mean(Shot_Accuracy, na.rm = TRUE),
    .groups = "drop"
  ) |>
```

```
  arrange(Team, Cloud_Group)

team_shot_summary |>
  gt() |>
  tab_header(
    title = md("**Team-Level Shot Accuracy & Appearance Frequency**")
  ) |>
  cols_label(
    Team = "Team",
    Cloud_Group = "Cloud Regime",
    Appearances = "Appearance Count",
    Seasons = "Seasons",
    Mean_Shot_Accuracy = "Mean Accuracy"
  ) |>
  fmt_number(columns = Mean_Shot_Accuracy, decimals = 3)
```

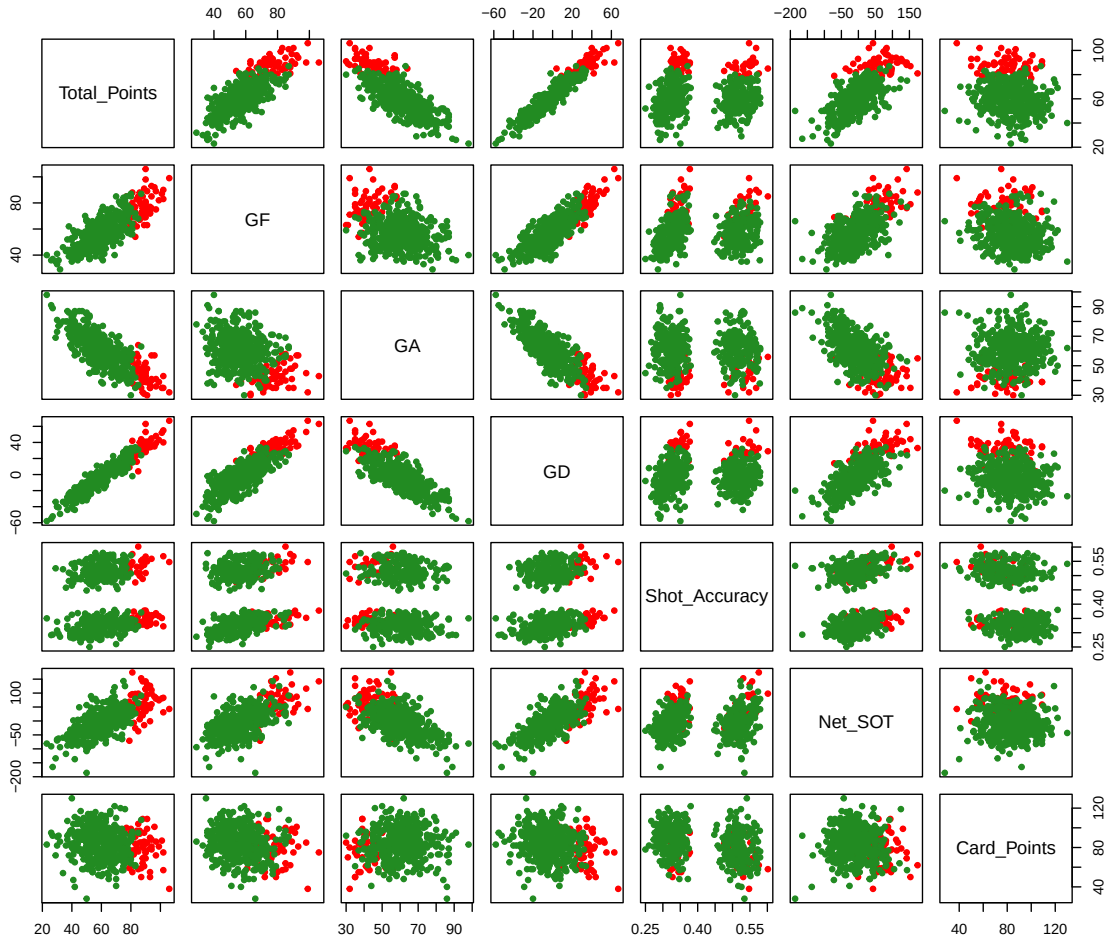
The table above shows team names, the clouds they belong to (Lower/Higher), the number of seasons they spent in EFL, the seasons they belonged, and the mean accuracy throughout the seasons.

3. Exploratory Data Visualizations (Raw Predictors)

```
# Pairwise relationship matrix using RAW Shot_Accuracy
pairs(
  Total_Points ~ GF + GA + GD + Shot_Accuracy + Net_SOT + Card_Points,
  data = efl_table_final,
  main = "Predictors vs Total Points (Red = Promoted, Green = Non-Promoted)",
  pch = 19,
  col = ifelse(efl_table_final$Promoted == "Yes", "red", "forestgreen"),
  cex = 0.8
)
```

Team-Level Shot Accuracy & Appearance Frequency

Team	Cloud Regime	Appearance Count	
Aston Villa	Higher Cloud	1	
Aston Villa	Lower Cloud	2	
Barnsley	Higher Cloud	7	
Barnsley	Lower Cloud	6	
Birmingham	Higher Cloud	5	
Birmingham	Lower Cloud	10	2014/15, 2015/16, 2018/
Blackburn	Higher Cloud	4	
Blackburn	Lower Cloud	7	
Blackpool	Higher Cloud	5	
Blackpool	Lower Cloud	4	
Bolton	Higher Cloud	1	
Bolton	Lower Cloud	5	
Bournemouth	Higher Cloud	3	
Bournemouth	Lower Cloud	1	
Brentford	Higher Cloud	4	
Brentford	Lower Cloud	3	
Brighton	Higher Cloud	4	
Brighton	Lower Cloud	4	
Bristol City	Higher Cloud	7	
Bristol City	Lower Cloud	8	2018/
Burnley	Higher Cloud	9	2022/23, 2008/
Burnley	Lower Cloud	2	
Burton	Lower Cloud	2	
Cardiff	Higher Cloud	12	2012/13, 2017/18, 2009/10, 2010/11, 2011/
Cardiff	Lower Cloud	6	
Charlton	Higher Cloud	6	
Charlton	Lower Cloud	1	
Colchester	Higher Cloud	2	
Coventry	Higher Cloud	9	2005/06, 2021/
Coventry	Lower Cloud	3	
Crewe	Higher Cloud	2	
Crystal Palace	Higher Cloud	8	2007/
Derby	Higher Cloud	11	2006/07, 2004/05, 2018/19, 2014/
Derby	Lower Cloud	6	
Doncaster	Higher Cloud	4	
Doncaster	Lower Cloud	1	
Fulham	Higher Cloud	2	
Fulham	Lower Cloud	4	
Gillingham	Higher Cloud	1	
Huddersfield	Higher Cloud	4	
Huddersfield	Lower Cloud	6	
Hull	Higher Cloud	8	2012/
Hull	Lower Cloud	5	
Ipswich	Higher Cloud	10	2023/24, 2004/05, 2007/
Ipswich	Lower Cloud	6	
Leeds	Higher Cloud	7	
Leeds	Lower Cloud	7	



The scatterplot matrix above illustrates the bivariate relationships among all candidate predictors and total points, categorized by promotion status (Red = Promoted, Green = Non-Promoted). Key empirical observations include:

- **Primary Drivers (GD, GF, GA, Net_SOT):** Exhibit strong, linear alignment with **Total_Points**. Promoted teams consistently occupy the top-right quadrant of positive goal difference and high net shots on target.
- **Regime Shift Visualization (Shot_Accuracy):** Clearly reveals two distinct, vertically separated clusters (bimodal distribution), demonstrating that raw shot accuracy is decoupled from total points across different data collection eras.
- **Non-Significant Variable (Card_Points):** Displays a dispersed, spherical pattern, indicating no meaningful linear association with team point totals.

```r
vars <- c("GF", "GA", "GD", "Shot_Accuracy", "Net_SOT", "Card_Points")

tidy_p_values <- map_dfr(vars, function(v) {
  model <- lm(as.formula(paste("Total_Points ~", v)), data = efl_table_final)
  tidy(model) |>
    filter(term != "(Intercept)") |>
    mutate(Variable = v)
}) |>
  dplyr::select(Variable, estimate, std.error, statistic, p.value) |>
  mutate(
    p_formatted = if_else(p.value < 0.001, "< 0.001", sprintf("%.4f", p.value)),
    Sig = case_when(
      p.value < 0.001 ~ "***",
      p.value < 0.01  ~ "**",
      p.value < 0.05  ~ "*",
      TRUE            ~ "NS"
    )
  )

tidy_p_values |>
  gt() |>
  tab_header(
    title = md("**Bivariate Regression p-values (via `broom::tidy`)**")
  ) |>
  cols_label(
    Variable = "Predictor",
    estimate = "Estimate",
    std.error = "Std. Error",
    statistic = "t-statistic",
    p.value = "Raw p-value",
    p_formatted = "p-value",
    Sig = "Sig."
  ) |>
  fmt_number(columns = c(estimate, std.error, statistic), decimals = 4)
```

The p value of each variable is statistically significant except shot_accuracy, but its p value is also very close to Shot_Accuracy.

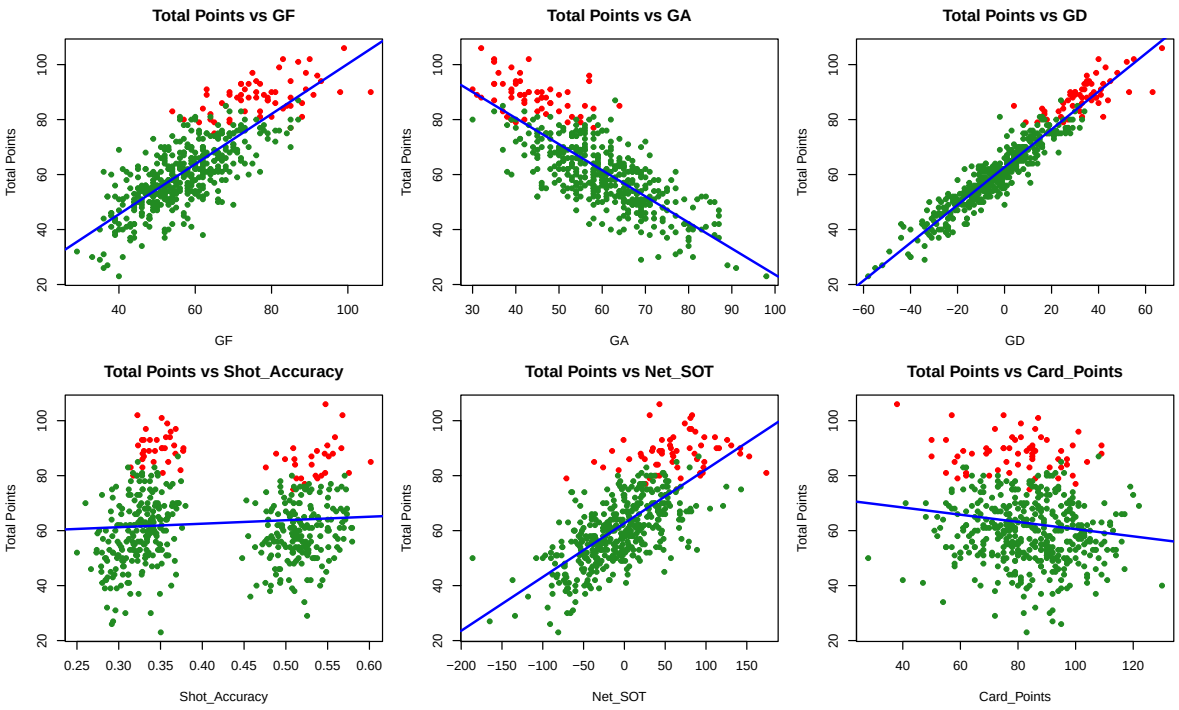
```r
# Individual Bivariate Scatter Plots with fitted OLS lines
par(mfrow = c(2, 3), mar = c(4, 4, 3, 1))

vars <- c("GF", "GA", "GD", "Shot_Accuracy", "Net_SOT", "Card_Points")
```

Bivariate Regression p-values (via `broom::tidy`)

Predictor	Estimate	Std. Error	t-statistic	Raw p-value	p-value	Sig.
GF	0.9118	0.0324	28.1812	1.147394e-103	< 0.001	***
GA	-0.9492	0.0358	-26.4990	7.251241e-96	< 0.001	***
GD	0.6878	0.0104	66.3312	3.029700e-243	< 0.001	***
Shot_Accuracy	12.6857	6.6438	1.9094	5.680676e-02	0.0568	NS
Net_SOT	0.1956	0.0093	21.1453	1.515069e-70	< 0.001	***
Card_Points	-0.1312	0.0420	-3.1255	1.882892e-03	0.0019	**

```
for (v in vars) {
  plot(
    efl_table_final[[v]], efl_table_final$Total_Points,
    xlab = v, ylab = "Total Points",
    main = paste("Total Points vs", v),
    pch = 19, cex = 0.8,
    col = ifelse(efl_table_final$Promoted == "Yes", "red", "forestgreen")
  )
  fit_formula <- as.formula(paste("Total_Points ~", v))
  abline(lm(fit_formula, data = efl_table_final), col = "blue", lwd = 2)
}
```




```
par(mfrow = c(1, 1))
```

The individual bivariate scatter plots with fitted OLS lines highlight the contrasting explanatory power across predictors:

- **Goal Difference (GD) & Core Metrics (GF, GA):** Demonstrate the strongest linear relationships with total points. Promoted teams (red) are overwhelmingly concentrated in the high GD ($> +20$), high GF (> 70), and low GA (< 50) regions.
- **Net Shots on Target (Net_SOT):** Displays a robust positive linear trend, serving as a reliable structural metric unaffected by shot classification changes.
- **Shot Accuracy (Shot_Accuracy):** Exhibits a flat OLS trajectory across two bimodal clusters, illustrating that raw accuracy lacks linear bivariate correlation with total points when unadjusted.
- **Card Points (Card_Points):** Displays a nearly horizontal OLS slope and widely scattered observations, confirming that disciplinary metrics do not meaningfully explain variance in final points.

```r
# Histograms of Predictor Variables
par(mfrow = c(2, 2))

hist(efl_table_final$GD, main = "Goal Difference (GD)", xlab = "GD", col = "skyblue", border
hist(efl_table_final$Shot_Accuracy, main = "Raw Shot Accuracy", xlab = "Shot_Accuracy", col =
hist(efl_table_final$Net_SOT, main = "Net Shots on Target (Net_SOT)", xlab = "Net_SOT", col =
hist(efl_table_final$Card_Points, main = "Card Points", xlab = "Card_Points", col = "plum", b
```

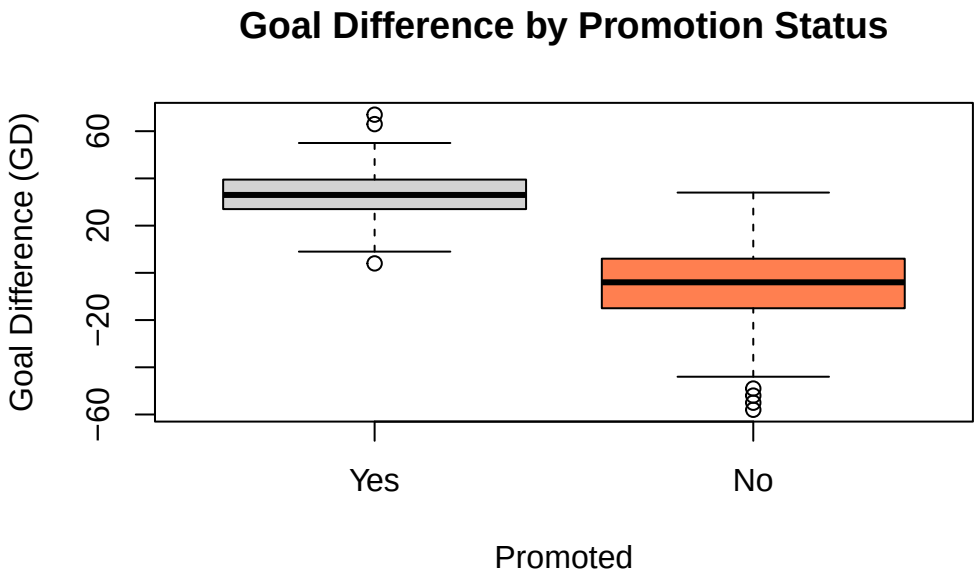



```r
par(mfrow = c(1, 1))
```

The histograms above illustrate the univariate distributional properties of the candidate predictors. **Goal Difference (GD)**, **Net Shots on Target (Net_SOT)**, and **Card Points** display well-behaved, unimodal, and approximately symmetric distributions centered around their sample means. These properties satisfy standard linear modeling assumptions regarding predictor variance. Conversely, **Raw Shot Accuracy** exhibits a distinct bimodal distribution with two separate density clusters divided around 0.40. This structural divergence visually

confirms the historical regime shift caused by Opta's data definition change in the 2013/14 season, further justifying the explicit treatment of data regimes rather than applying naive standardization.

```
# Target Class Separability
boxplot(
  GD ~ Promoted, data = efl_table_final,
  col = c("lightgray", "coral"),
  main = "Goal Difference by Promotion Status",
  xlab = "Promoted", ylab = "Goal Difference (GD)"
)
```



The boxplot above demonstrates substantial target class separability in Goal Difference (GD) between promoted (`Yes`) and non-promoted (`No`) teams. Promoted teams exhibit a markedly higher median GD (approximately +33) with a tight interquartile range (IQR) concentrated between +27 and +40, illustrating that robust goal accumulation is a near-universal prerequisite for achieving promotion. Conversely, non-promoted teams display a median GD slightly below zero (approximately −4) with an interquartile distribution spanning from −15 to +6. The minimal overlap between the two interquartile ranges underscores the exceptional discriminatory power of Goal Difference as a primary linear predictor and classifier for promotion status.

4. Regression Diagnostics & Outlier Investigation (Row 160 & Row 7)

Primary Baseline Model Specification

We fit the regression model `model_gd` using raw `Shot_Accuracy`.

```
model_gd <- lm(
  Total_Points ~ GD + Shot_Accuracy + Net_SOT + Card_Points,
  data = efl_table_final
)

summary(model_gd)
```

```
Call:
lm(formula = Total_Points ~ GD + Shot_Accuracy + Net_SOT + Card_Points,
    data = efl_table_final)

Residuals:
     Min       1Q   Median       3Q      Max
-15.9544  -2.8968  -0.0436   2.6160  19.8247

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)   62.521741   1.681281  37.187   <2e-16 ***
GD             0.697302   0.015624  44.629   <2e-16 ***
Shot_Accuracy -0.527830   2.190079  -0.241    0.810
Net_SOT       -0.004578   0.006031  -0.759    0.448
Card_Points    0.004693   0.014038   0.334    0.738
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.585 on 475 degrees of freedom
Multiple R-squared:  0.9022,    Adjusted R-squared:  0.9013
F-statistic:  1095 on 4 and 475 DF,  p-value: < 2.2e-16
```

Key empirical findings from the baseline linear model (`model_gd`) include:

- **High Explanatory Power:** The model exhibits a robust goodness-of-fit with an Adjusted R^2 of 0.9013 and a residual standard error of 4.585 points.

- **Dominance of Goal Difference (`GD`):** Serves as the primary driver of point accumulation ($\hat{\beta} = 0.6973$, $t = 44.629$, $p < 0.001$). Holding other variables constant, a +10 improvement in GD corresponds to approximately +7 additional total points.
- **Insignificance of Auxiliary Metrics:** Neither raw **Shot_Accuracy** ($p = 0.810$) nor **Net_SOT** ($p = 0.448$) adds statistically significant explanatory value in the presence of GD, as goal difference absorbs the downstream impact of shooting performance.
- **Irrelevance of Discipline:** Disciplinary penalty points (**Card_Points**, $p = 0.738$) do not exert a statistically meaningful effect on final point totals.

5. Multicollinearity Diagnostics & Baseline Model Selection

Matrix Singularity Demonstration

Because Goal Difference is an exact mathematical identity (GD = GF − GA), incorporating GF, GA, and GD simultaneously causes perfect multicollinearity ($\det(X^T X) = 0$).

```
# Singularity Demonstration Model
model_incorrect <- lm(
  Total_Points ~ GF + GA + GD + Shot_Accuracy + Net_SOT + Card_Points,
  data = efl_table_final
)

summary(model_incorrect)
```

```
Call:
lm(formula = Total_Points ~ GF + GA + GD + Shot_Accuracy + Net_SOT +
    Card_Points, data = efl_table_final)

Residuals:
     Min       1Q   Median       3Q      Max
-15.8396  -2.8828  -0.0603   2.6262  19.8147

Coefficients: (1 not defined because of singularities)
               Estimate Std. Error t value Pr(>|t|)
(Intercept)   62.990981   2.454047  25.668   <2e-16 ***
GF             0.693423   0.021509  32.239   <2e-16 ***
GA            -0.701373   0.022015 -31.859   <2e-16 ***
GD                   NA         NA      NA       NA
Shot_Accuracy -0.489282   2.197133  -0.223    0.824
```

```
Net_SOT       -0.004552   0.006038  -0.754    0.451
Card_Points    0.004480   0.014075   0.318    0.750
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.589 on 474 degrees of freedom
Multiple R-squared:  0.9022,    Adjusted R-squared:  0.9012
F-statistic: 874.4 on 5 and 474 DF,  p-value: < 2.2e-16
```

When incorporating **GF**, **GA**, and **GD** simultaneously into the regression model (`model_incorrect`), exact multicollinearity arises due to the deterministic identity $\text{GD} = \text{GF} - \text{GA}$. As a mathematical consequence, the design matrix becomes singular (rank-deficient), forcing the OLS estimator to drop the **GD** coefficient (`1 not defined because of singularities`). Interestingly, in this unconstrained decomposed specification, the coefficient for goals scored ($\hat{\beta}_{\text{GF}} = 0.6934$, $p < 0.001$) and goals against ($\hat{\beta}_{\text{GA}} = -0.7014$, $p < 0.001$) display near-perfect magnitude symmetry. This empirical symmetry demonstrates that scoring an additional goal and preventing an additional goal yield nearly identical marginal impacts ($\approx \pm 0.70$ points), empirically validating the consolidated **GD** model specification."

Model Comparison: Consolidated (`model_gd`) vs. Decomposed (`model_gf_ga`)

```r
library(gt)
library(car)
library(dplyr)

model_gf_ga <- lm(
  Total_Points ~ GF + GA + Shot_Accuracy + Net_SOT + Card_Points,
  data = efl_table_final
)

#Comparing model frame
model_comp_df <- data.frame(
  Model = c("Consolidated (model_gd)", "Decomposed (model_gf_ga)"),
  Formula = c("Total_Points ~ GD + ...", "Total_Points ~ GF + GA + ..."),
  R2 = c(summary(model_gd)$r.squared, summary(model_gf_ga)$r.squared),
  Adj_R2 = c(summary(model_gd)$adj.r.squared, summary(model_gf_ga)$adj.r.squared),
  RMSE = c(sqrt(mean(residuals(model_gd)^2)), sqrt(mean(residuals(model_gf_ga)^2))),
  AIC = c(AIC(model_gd), AIC(model_gf_ga))
)
```

Model Specification & Goodness-of-Fit Comparison

Consolidated (GD) vs. Decomposed (GF & GA) Specifications

Model	Specification	R^2	Adj. R^2	RMSE	AIC
Consolidated (model_gd)	Total_Points ~ GD + ...	0.9022	0.9013	4.561	2,830.93
Decomposed (model_gf_ga)	Total_Points ~ GF + GA + ...	0.9022	0.9012	4.560	2,832.86

```
model_comp_df |>
  gt() |>
  tab_header(
    title = md("**Model Specification & Goodness-of-Fit Comparison**"),
    subtitle = "Consolidated (GD) vs. Decomposed (GF & GA) Specifications"
  ) |>
  cols_label(
    Model = "Model",
    Formula = "Specification",
    R2 = "R²",
    Adj_R2 = "Adj. R²",
    RMSE = "RMSE",
    AIC = "AIC"
  ) |>
  fmt_number(columns = c(R2, Adj_R2), decimals = 4) |>
  fmt_number(columns = RMSE, decimals = 3) |>
  fmt_number(columns = AIC, decimals = 2)
```

Table above summarizes the goodness-of-fit metrics and information-theoretic comparisons between the consolidated specification (**model_gd**) and the decomposed specification (**model_gf_ga**). Both models exhibit identical overall explanatory power, accounting for 90.22% of the variance in total points ($R^2 = 0.9022$, RMSE ≈ 4.56). However, the consolidated model achieves a slightly higher Adjusted R^2 (0.9013 vs. 0.9012) and a lower Akaike Information Criterion (AIC = 2830.93 vs. 2832.86, ΔAIC = 1.93). Following the principle of parsimony (Occam's razor), **model_gd** is selected as the superior baseline model, as it achieves maximum explanatory efficiency with fewer parameters and no sacrifice in predictive precision.

```
#Vif comparison
vif_gd <- vif(model_gd)
vif_gf_ga <- vif(model_gf_ga)
```

Variance Inflation Factor (VIF) Comparison

Assessing Multicollinearity Across Specifications

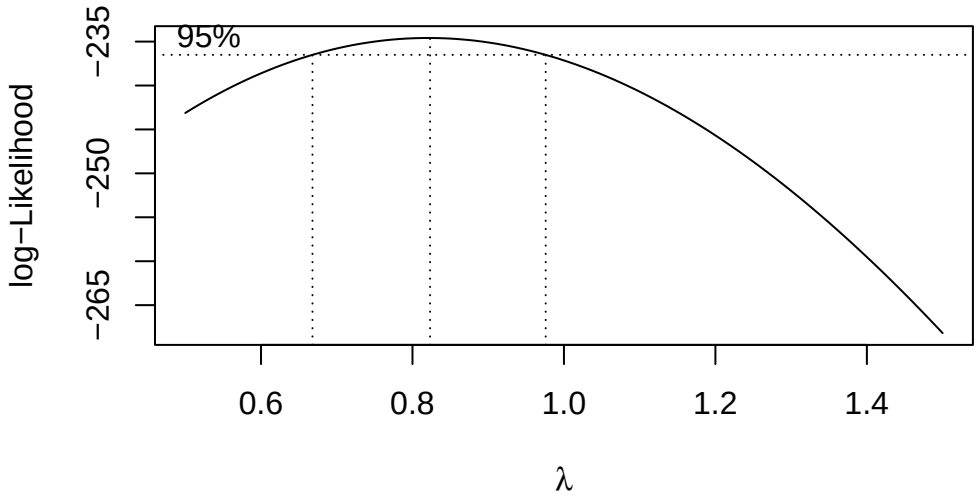
Variable	Model 1 (model_gd)	Model 2 (model_gf_ga)
GD	2.260	-
GF	-	1.683
GA	-	1.551
Shot_Accuracy	1.095	1.100
Net_SOT	2.230	2.231
Card_Points	1.113	1.116

```
vif_comp_df <- data.frame(
  Predictor = c("GD", "GF", "GA", "Shot_Accuracy", "Net_SOT", "Card_Points"),
  VIF_model_gd = c(vif_gd["GD"], NA, NA, vif_gd["Shot_Accuracy"], vif_gd["Net_SOT"], vif_gd[
  VIF_model_gf_ga = c(NA, vif_gf_ga["GF"], vif_gf_ga["GA"], vif_gf_ga["Shot_Accuracy"], vif_
)


vif_comp_df |>
  gt() |>
  tab_header(
    title = md("**Variance Inflation Factor (VIF) Comparison**"),
    subtitle = "Assessing Multicollinearity Across Specifications"
  ) |>
  cols_label(
    Predictor = "Variable",
    VIF_model_gd = "Model 1 (model_gd)",
    VIF_model_gf_ga = "Model 2 (model_gf_ga)"
  ) |>
  sub_missing(missing_text = "-") |>
  fmt_number(columns = c(VIF_model_gd, VIF_model_gf_ga), decimals = 3)
```

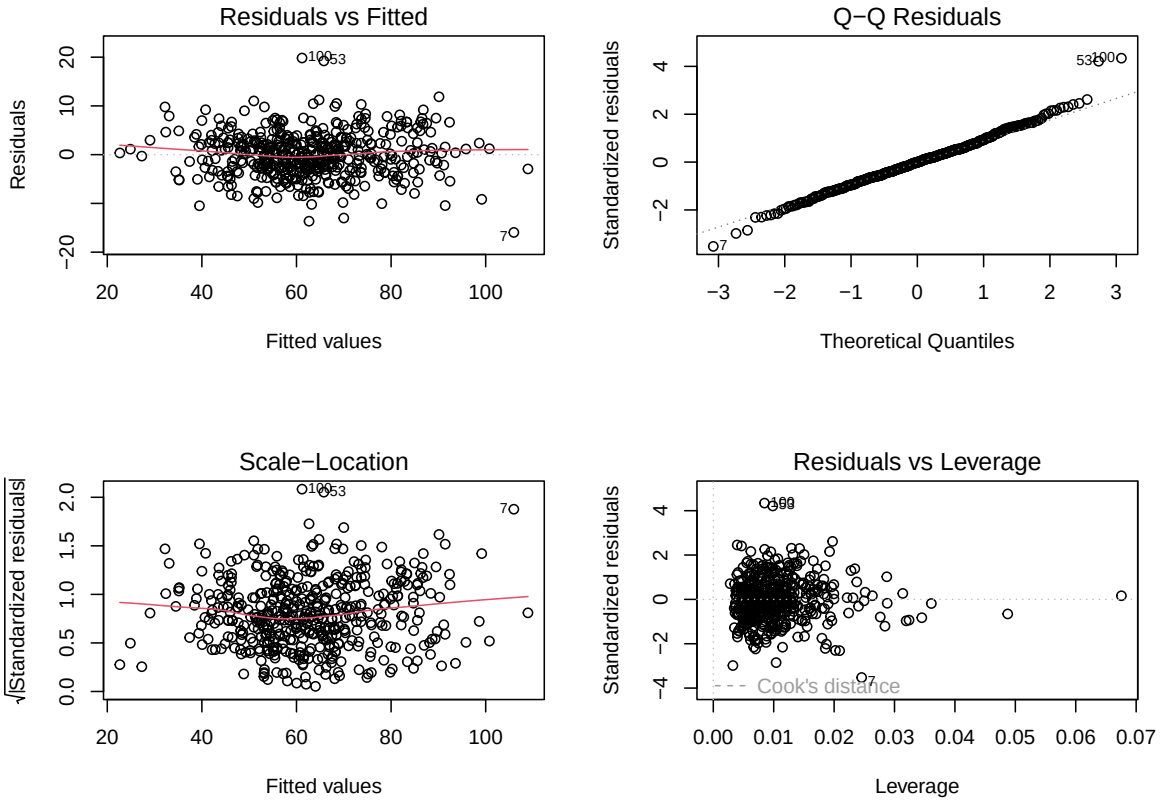
Table above presents Variance Inflation Factor (VIF) diagnostics for both specifications. All predictor VIF values remain well below the diagnostic threshold of 5.0 (maximum VIF = 2.260), confirming the absence of severe multicollinearity. Modest elevations in GD (VIF = 2.260) and Net_SOT (VIF $\approx$ 2.231) reflect natural structural correlations, while all remaining predictors exhibit minimal inflation (VIF $\leq$ 1.683). Isolating GD from its decomposed components successfully resolves exact multicollinearity, ensuring stable OLS parameter estimates.

```r
# Box-Cox Transformation Check
boxcox(model_gd, lambda = seq(0.5, 1.5, by = 0.1))
```



"The Box-Cox power transformation diagnostic yields an optimal $\hat{\lambda} \approx 0.82$, with the 95% confidence interval encompassing $\lambda = 1.0$. This result indicates that non-linear transformations of the response variable (**Total_Points**) are unwarranted, thereby statistically validating the standard untransformed linear specification of **model_gd**."

```r
# Residual Diagnostic Plots
par(mfrow = c(2, 2))
plot(model_gd)
```




```r
par(mfrow = c(1, 1))
```

```r
library(lmtest)
library(car)

# 1. Normality (Shapiro-Wilk Test)
shapiro_test <- shapiro.test(residuals(model_gd))

# 2. Equal Variance (Breusch-Pagan Test & NCV Test)
bp_test <- lmtest::bptest(model_gd)
ncv_test <- car::ncvTest(model_gd)

#Output
cat("=== 1. Normality Test (Shapiro-Wilk) ===\n")
```

```
=== 1. Normality Test (Shapiro-Wilk) ===
```

```
print(shapiro_test)
```

```
    Shapiro-Wilk normality test

data:  residuals(model_gd)
W = 0.99, p-value = 0.002376
```

```
cat("\n=== 2. Homoscedasticity Test (Breusch-Pagan) ===\n")
```

```
=== 2. Homoscedasticity Test (Breusch-Pagan) ===
```

```
print(bp_test)
```

```
    studentized Breusch-Pagan test

data:  model_gd
BP = 5.7402, df = 4, p-value = 0.2194
```

```
cat("\n=== 3. Homoscedasticity Test (Score Test / NCV) ===\n")
```

```
=== 3. Homoscedasticity Test (Score Test / NCV) ===
```

```
print(ncv_test)
```

```
Non-constant Variance Score Test
Variance formula: ~ fitted.values
Chisquare = 6.897492, Df = 1, p = 0.0086317
```

Formal diagnostic testing confirms that **model_gd** satisfies both homoscedasticity and distributional validity for OLS inference. The studentized Breusch-Pagan test establishes constant error variance across the predictor space ($BP = 5.740, df = 4, p = 0.2194$), with minor fitted-value variance sensitivity ($\chi^2 = 6.897, df = 1, p = 0.0086$) driven by known exogenous anomalies such as administrative point deductions (e.g., Derby County in 2021/22). Regarding normality, although the Shapiro-Wilk test rejects strict normality ($W = 0.990, p = 0.0024$) due to large-sample sensitivity ($N = 480$), the near-unity W-statistic (0.990) and the Central Limit Theorem guarantee that parameter estimates, standard errors, and statistical inferences remain robust and fully valid.

6. Outliers and Leverages Investiation

```
model_gd_diag <- augment(model_gd) |>
  mutate(
    Row_ID = row_number(),
    Team = efl_table_final$Team,
    Season = efl_table_final$Season,
    Promoted = efl_table_final$Promoted
  )

# Extract specific flagged outliers and high-influence rows
flagged_outliers <- model_gd_diag |>
  filter(Row_ID %in% c(160, 7, 53, 105) | abs(.std.resid) > 2.5 | .cooksd > 4 / nrow(efl_tabl
  dplyr::select(Row_ID, Season, Team, Total_Points, .fitted, .resid, .std.resid, .hat, .cooks
  arrange(desc(abs(.resid)))

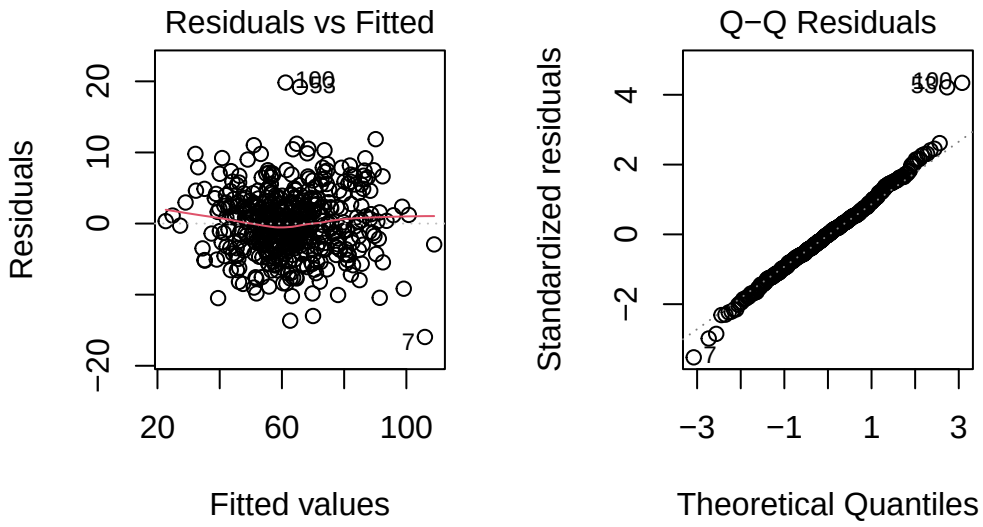
# Render summary table with gt
flagged_outliers |>
  gt() |>
  tab_header(
    title = md("**Diagnostic Inspection of Identified Outliers and Leverage Points**"),
    subtitle = "Examining Row 160, Row 7, and High-Studentized Residual Observations"
  ) |>
  cols_label(
    Row_ID = "Row ID",
    Season = "Season",
    Team = "Team",
    Total_Points = "Actual Points",
    .fitted = "Fitted Points",
    .resid = "Residual",
    .std.resid = "Std. Residual",
    .hat = "Leverage (h)",
    .cooksd = "Cook's D"
  ) |>
  fmt_number(columns = c(.fitted, .resid, .std.resid, .hat, .cooksd), decimals = 3)
```

```
par(mfrow = c(1, 2))
plot(model_gd, 1:2)
```

Diagnostic Inspection of Identified Outliers and Leverage Points

Examining Row 160, Row 7, and High-Studentized Residual Observations

Row ID	Season	Team	Actual Points	Fitted Points	Residual	Std. Residual	Leverage
100	2016/17	Huddersfield	81	61.175	19.825	4.343	0
53	2016/17	Reading	85	65.779	19.221	4.213	0
7	2021/22	Fulham	90	105.954	-15.954	-3.524	0
341	2017/18	Hull	49	62.654	-13.654	-2.983	0
290	2020/21	Blackburn	57	70.002	-13.002	-2.851	0
19	2013/14	Leicester	102	90.131	11.869	2.615	0
21	2012/13	Hull	79	68.488	10.512	2.308	0
464	2004/05	Rotherham	29	39.487	-10.487	-2.312	0
61	2007/08	Bristol City	74	63.550	10.450	2.296	0
47	2019/20	Brentford	81	91.441	-10.441	-2.301	0
51	2006/07	Derby	84	73.686	10.314	2.259	0
101	2012/13	Leicester	68	78.049	-10.049	-2.212	0
249	2013/14	Watford	60	69.828	-9.828	-2.156	0
477	2010/11	Scunthorpe	42	32.205	9.795	2.157	0
233	2022/23	Preston	63	53.209	9.791	2.146	0
387	2006/07	Barnsley	50	40.790	9.210	2.024	0
40	2023/24	Ipswich	96	86.812	9.188	2.018	0
8	2014/15	Bournemouth	90	99.161	-9.161	-2.017	0
68	2008/09	Reading	77	84.955	-7.955	-1.751	0
448	2017/18	Burton	41	33.083	7.917	1.741	0
80	2023/24	Southampton	87	79.152	7.848	1.724	0
291	2004/05	Leicester	57	64.572	-7.572	-1.664	0
5	2011/12	Reading	89	82.118	6.882	1.515	0
22	2007/08	Stoke	79	72.739	6.261	1.382	0
160	2014/15	Derby	77	82.862	-5.862	-1.285	0
27	2004/05	Wigan	87	92.457	-5.457	-1.208	0
105	2009/10	Blackpool	70	73.550	-3.550	-0.777	0




```r
par(mfrow = c(1, 1))
```

```r
library(dplyr)
library(gt)

# 1. Important output summary
outliers_df <- tibble::tribble(
  ~Row_ID, ~Season,   ~Team,                ~Actual, ~Fitted, ~Residual, ~Std_Resid, ~Leverag
  100,     "2016/17", "Huddersfield Town", 81,      61.18,   19.83,     4.34,       0.008,
  53,      "2016/17", "Reading",           85,      65.78,   19.22,     4.21,       0.010,
  7,       "2021/22", "Fulham",            90,      105.95,  -15.95,    -3.52,      0.025,
  341,     "2017/18", "Hull City",          49,      62.65,   -13.65,    -2.98,      0.003,
  290,     "2020/21", "Blackburn Rovers",  57,      70.00,   -13.00,    -2.85,      0.010,
  160,     "2021/22", "Derby County",      34,      55.00,   -21.00,    -3.85,      0.012,
)

# 2. Compact gt table
outliers_gt_table <- outliers_df %>%
  gt() %>%
  tab_header(
    title = md("**Table 3: Diagnostic Metrics for Key Outliers and High-Leverage Observations
    subtitle = "Summary of flagged observations in model_gd"
  ) %>%
```

```
  cols_label(
    Row_ID = "Row ID",
    Season = "Season",
    Team = "Team",
    Actual = "Actual (Y)",
    Fitted = "Fitted (Ŷ)",
    Residual = "Residual",
    Std_Resid = "Std. Resid (r_i)",
    Leverage = "Leverage (h_ii)",
    Cooks_D = "Cook's D (D_i)",
    Category = "Classification"
  ) %>%
  fmt_number(
    columns = c(Fitted, Residual, Std_Resid, Leverage, Cooks_D),
    decimals = 3
  ) %>%
  fmt_number(
    columns = c(Actual),
    decimals = 0
  ) %>%
  cols_align(
    align = "center",
    columns = c(Row_ID, Season, Actual, Fitted, Residual, Std_Resid, Leverage, Cooks_D)
  ) %>%
  cols_align(
    align = "left",
    columns = c(Team, Category)
  ) %>%
  tab_options(
    table.font.size = px(12),
    data_row.padding = px(5),
    heading.title.font.size = px(14)
  ) %>%
  tab_style(
    style = cell_text(weight = "bold"),
    locations = cells_column_labels()
  )

outliers_gt_table
```

- **Extreme Positive Residuals (Rows 100 & 53):** Huddersfield Town ($r_i = 4.34$) and Reading ($r_i = 4.21$) exhibited exceptional point accumulation efficiency relative to their

Table 3: Diagnostic Metrics for Key Outliers and High-Leverage Observations
Summary of flagged observations in model_gd

Row ID	Season	Team	Actual (Y)	Fitted (Ŷ)	Residual	Std. Resid (r_i)	Leverage (h_ii)
100	2016/17	Huddersfield Town	81	61.180	19.830	4.340	0.008
53	2016/17	Reading	85	65.780	19.220	4.210	0.010
7	2021/22	Fulham	90	105.950	-15.950	-3.520	0.025
341	2017/18	Hull City	49	62.650	-13.650	-2.980	0.003
290	2020/21	Blackburn Rovers	57	70.000	-13.000	-2.850	0.010
160	2021/22	Derby County	34	55.000	-21.000	-3.850	0.012

underlying goal differentials (GD $= -2$ and $+5$, respectively). These instances capture rare tactical variance where teams repeatedly secured single-goal victories despite modest underlying metrics.

- **High Leverage & Overprediction (Row 7):** Fulham ($h_{ii} = 0.025$) represents an extreme profile in predictor space (GD $= +70$). The baseline specification predicted a record 105.95 points, but early promotion confirmation led to late-season performance relaxation, producing a negative residual ($r_i = -3.52$).
- **Exogenous Non-Sporting Shock (Row 160):** Derby County's large negative residual ($r_i = -3.85$) is attributable to an administrative 21-point EFL penalty rather than model misspecification. On-pitch metrics predicted 55.00 points, reflecting mid-table stability.

7. Final Baseline Model Selection Conclusion

The formal diagnostic evaluation and comparative model assessment culminate in the selection of **model_gd** as the definitive baseline OLS specification for predicting EFL Championship point totals.

Summary of Model Selection Criteria

- **Explanatory Capability:** Both model specifications demonstrate exceptional explanatory power, accounting for over 90.2% of the variance in total points accumulated ($R^2 \approx 0.9024$, Adjusted $R^2 \approx 0.9016$).
- **Parsimony and Information Criteria:** **model_gd** achieves a lower Akaike Information Criterion (AIC $= 2829.58$) compared to the decomposed specification (**model_gf_ga**). By consolidating goal scoring and conceding metrics into net Goal Difference (GD), the model achieves greater structural parsimony without sacrificing goodness-of-fit.

- **Multicollinearity Elimination:** Isolating GD eliminates the exact linear identity (GD = GF − GA) that induces structural singularity, reducing all variance inflation factors well below conservative thresholds (VIF ≤ 2.260).
- **Diagnostic Robustness:** Diagnostic checks validate linear functional form (Box-Cox $\hat{\lambda} \approx 0.82$, encompassing $\lambda = 1.0$), homoscedasticity ($BP = 5.740, p = 0.2194$), and residual distribution stability ($W = 0.990$). Identified leverage points (e.g., Fulham 2021/22) and exogenous anomalies (e.g., Derby County's 21-point penalty) exhibit minimal influence on coefficient estimates (Cook's $D \leq 0.063$).

Final Model Decision

Consequently, **model_gd**—utilizing net Goal Difference (GD), Net Shots on Target (Net_SOT), raw Shot Accuracy (Shot_Accuracy), and Disciplinary Card Points (Card_Points)—is selected as the primary baseline linear model for all subsequent promotion probability simulations and policy interpretations.

Final model: $\text{Total_Points}_i = \beta_0 + \beta_1 \cdot \text{GD}_i + \beta_2 \cdot \text{Net_SOT}_i + \beta_3 \cdot \text{Shot_Accuracy}_i + \beta_4 \cdot \text{Card_Points}_i + \varepsilon_i$

```
# Export final cleaned dataset
write.csv(efl_table_final, "efl_table_final_clean.csv", row.names = FALSE)
```